

10 Central Force Motion: The Kepler Problem

The orbital motion of a planet around the Sun was one of the first important problems to be analyzed in terms of Newton's laws of motion. The gravitational force attracting a planet to the Sun is a *central force*, that is, a force directed towards a fixed point. The motion of a planet is a prime example of the more general problem of the behavior of a particle acted upon by a central force.

Although we shall be primarily concerned with the motion of planets and satellites, the techniques you will learn in this chapter are applicable to any kind of central force. In this chapter, as well as learning the laws governing the motion of celestial bodies, you will be exposed to the concept of *effective potential energy*, and how constants of the motion are used in solving physics problems.

Historically, the quantitative analysis of orbital motion began with Kepler's realization that the motion of planets can be described by three empirical laws. An important point made in this chapter is that Kepler's laws of planetary motion can be derived theoretically from Newton's laws of motion. Additionally, Newton's laws give us a much deeper understanding of Kepler's laws. This application of Newton's ideas amazed and fascinated the "natural philosophers" of his era and was one of the most important events in the history of science.¹

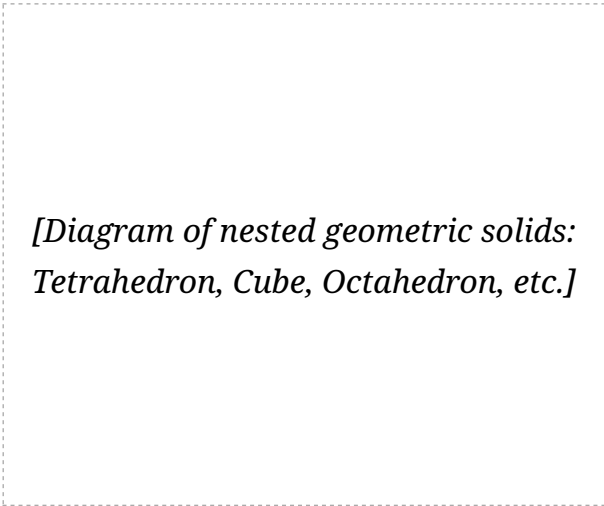
10.1 Johannes Kepler (Historical Note)

Johannes Kepler lived from 1571 to 1630. He was born into a poor family in a small town in Germany. His father was a professional soldier who spent much of his time away from home and his mother was a quarrelsome woman who was accused of witchcraft in her old age. It would be fair to assume that Kepler had an unhappy childhood. Nevertheless, he was very intelligent and an excellent student. The prince who ruled the region sent him to study at a Lutheran seminary. Eventually, Kepler graduated from the University of Tübingen. Originally he planned to study divinity, but before he was ordained, the authorities in the seminary convinced him that he was not cut out for the clergy. With the help of his advisers, he was appointed to a

position teaching mathematics in Graz, Austria. He was a quiet, introspective person and not greatly interested in teaching; in fact, he was probably a terrible teacher for he had the reputation of interrupting his own lectures to silently mull over some idea that had just occurred to him.

In Graz, Kepler had an idea that changed the course of his life. This idea, which became an obsession with him, was (he thought) a glimpse into the mind of God: a vision of the basic structure of the universe.

¹ Much of the material in this chapter is an introduction to celestial mechanics. Many excellent celestial mechanics books are available, including A. E. Roy, *Orbital Motion*, Adam Hilger Press, Bristol, 1988, and J. M. A. Danby, *Fundamentals of Celestial Mechanics*, William-Bell Inc., Richmond, VA, 1992. A more recent book that is particularly well written is *Solar System Dynamics* by C. D. Murray and S. F. Dermott, Cambridge University Press, Cambridge, 1999.



*[Diagram of nested geometric solids:
Tetrahedron, Cube, Octahedron, etc.]*

**Figure 10.1 A nested tetrahedron and cube with inscribed and circumscribed spheres.
In Kepler's model the nested perfect solids (from innermost outwards) were an
octahedron, an icosahedron, a dodecahedron, a tetrahedron, and a cube.**

It was, so to speak, revealed to him why there are only five planets, and why they are in their particular orbits around the Sun. Kepler's mind-boggling inspiration was this: There are only five planets (besides Earth) because there are only five "perfect" solids, and the orbits of the planets correspond to spheres inscribed and circumscribed about the perfect solids when they are nested, one inside the other.

The “perfect” solids (or “simple polyhedrons”) are the geometrical figures formed from regular polygons. A regular polygon is one with equal sides. Thus, for example, a tetrahedron (or equilateral pyramid) is made up of four equilateral triangles. A cube is made up of six squares. Similarly, an octahedron is an eight-sided solid made of equilateral triangles, an icosahedron is made of twenty equilateral triangles, and the dodecahedron is composed of twelve pentagons. Many other solids can be constructed from polygons; for example, a “soccer ball” shape can be made up of pentagons and hexagons. There are, however, only *five* “perfect” solids whose faces are a *single* type of regular polygon. There is a very neat proof that there can *only* be five such solids; this proof can be traced back to the ancient Greeks. If you are interested, it is reproduced in the book *Cosmos*, by Carl Sagan.²

Kepler’s idea was that if the perfect solids were placed one inside the other, and each was circumscribed with a sphere, then the Sun would be at the center of the system and each of the five planets would orbit the Sun in a circular orbit whose radius would be equal to the radius of the corresponding circumscribed sphere. Kepler built models of these nested solids and their circumscribed spheres, but he could not prove his theory because he did not have enough information on the distances of the five (known) planets to the Sun. Figure 10.1 illustrates the idea behind Kepler’s model.

At that time, the best astronomical data in Europe were in the observatory of an eccentric Danish astronomer named Tycho Brahe.³ Kepler went to visit Brahe to get his data. He was certain that Brahe would be overwhelmed by his wonderful new theory. Brahe was not overwhelmed. In fact, at first Brahe would not even let Kepler see the data! However, Brahe soon realized that Kepler was an excellent mathematician and he offered to let him work on a small portion of the data to calculate the orbit of Mars. This was certainly not what Kepler had in mind, but he grudgingly agreed.

² Carl Sagan, *Cosmos*, Random House, New York, 1980 (Appendix 2).

³ Tycho Brahe (1546–1601) set up an observatory at Uraniborg on the Danish island of Hven. A stream of young assistants from all over Europe came to Uraniborg where they carried out experiments in chemistry during the day and observed the heavens at night. Uraniborg has been described as the first research institution involved in “big science.” For the full story read the book by John Robert Christianson, *On Tycho’s Island: Tycho Brahe and his Assistants, 1570–1601*, Cambridge University Press, Cambridge, 2000. At the time Kepler went to visit him, Brahe had moved to Prague.

This was, perhaps, the original “graduate student–professor” relationship in science. To this day the same pattern exists. A young, aspiring scientist makes the pilgrimage to the laboratory of the established professor with the hope of being allowed to share in the professor’s knowledge and data. You yourself may be doing this a few years from now. I hope your relationship with your graduate advisor will be less tempestuous than that of Kepler and Tycho. They did not get along at all. Tycho was a man who loved a party, who spent his evenings eating, drinking, and carousing, whereas Kepler was somber and rather puritanical and did not at all approve of Tycho’s lifestyle.

Some years after the collaboration began, Tycho died. (Rumor has it that he died of a burst bladder while on a drinking spree.) Kepler inherited all of Tycho’s data as well as Tycho’s position as court astronomer. After a great deal of analysis, much to his dismay, Kepler found that Tycho’s data did not support his grandiose theory. In fact, the data showed that the orbits of planets were not even circles, but rather ellipses! Kepler tried to modify his theory by slipping elliptical orbits between the inscribed perfect solids, but it did not quite work. Kepler never did know that there are more than five planets, as the discoveries of the planets Uranus and Neptune came many years after his death. By the time these were discovered, Kepler’s theory on the inscribed orbits of the planets was no more than a historical oddity.

Kepler’s life is full of instructive incidents for physicists. His most valuable contribution to science was his analysis and synthesis of the observations of Tycho Brahe. It is interesting to consider that he was obsessed by a beautiful theory that did not agree with experiment. No matter how beautiful a theory may be, if it does not agree with experimental measurements, it must be discarded! Physics is the study of the physical universe and it is Nature that determines the way things behave. It is to Kepler’s credit that he respected and believed the data, even though the data did not agree with his theory.

10.2 Kepler’s Laws

Kepler’s synthesis of Tycho’s planetary data can be expressed as three statements that are now called Kepler’s laws of planetary motion. They are:

1. The orbit of a planet is an ellipse with the Sun at one focus.
2. The radius vector of a planet (the Sun–planet line) sweeps out equal areas in equal times.
3. The period of a planet squared is proportional to the cube of the semimajor axis

of its elliptical orbit.

These laws express three facts about the motion of a planet. They are “empirical laws,” that is, they were obtained from the data but they had no theoretical basis. Kepler’s laws describe the motion of planets but neither Kepler nor anyone else living at that time could give a reasonable explanation for this behavior. About 70 years later, Newton applied the law of universal gravitation and his laws of motion to the problem and succeeded brilliantly in showing that Kepler’s laws are a consequence of some very basic physical relations. It is no wonder that other scientists of his era were in such awe of him. In this chapter, I will show you what Newton did, but of course I will use modern methods. Near the end of the chapter I will return to Kepler’s laws: by that time you should have a deeper appreciation for them.

10.3 Central Forces

Any force that is directed toward or away from a fixed point (often taken as the origin of coordinates) and whose magnitude is a function *only* of the distance to the fixed point is called a *central force*. In general, a central force will have the form

$$\mathbf{F} = f(r)\hat{\mathbf{r}},$$

where $f(r)$ is the magnitude of the force (a function only of r) and $\hat{\mathbf{r}}$ is the radial unit vector. The direction of the force is along the line joining the particle and the origin. Two important central forces are the gravitational force between masses and the electrostatic force between charges. Assuming one of the bodies is fixed at the origin, the gravitational force is a central force given by

$$\mathbf{F} = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}},$$

and the electrostatic force is

$$\mathbf{F} = \frac{Q_1Q_2}{4\pi\epsilon_0r^2}\hat{\mathbf{r}}.$$

We can also *imagine* other central forces such as, for example,

$$\mathbf{F} = \frac{k}{r^5} \hat{\mathbf{r}}.$$

Such forces may or may not exist in nature, but we can analyze them mathematically anyway. Although this seems like a useless theoretical exercise, such studies may have a practical outcome. For example, to a first approximation, the force between molecules can be expressed as the sum of two central forces. This “Lennard-Jones” force can be expressed in terms of the potential energy as

$$V(r) = -\frac{a}{r^6} + \frac{b}{r^{12}}.$$

Here a and b are constants that depend on the properties of the particular molecules involved in the interaction. The force itself is obtained from $\mathbf{F} = -\nabla V$.

We are interested in determining the motion of a particle subjected to the gravitational force of another body (which we also treat as if it were a particle). For the sake of simplicity let us assume the more massive body is at rest. Strictly speaking, this cannot be true; the two objects actually orbit about their center of mass. But if one particle is much more massive than the other, then the more massive particle has a much smaller acceleration and in the limit, as the ratio of the masses goes to infinity, the more massive particle can be considered to be at rest.

To appreciate this, imagine a binary star system in which both stars have the same mass. The two stars will orbit around their center of mass, which is at a point halfway between them. But suppose one of the stars is more massive than the other. The center of mass will be closer to the more massive star. If one star is infinitely more massive than the other, the center of mass will be at the center of the larger star. Similarly, the International Space Station and the Earth are orbiting around their center of mass, but for all intents and purposes, the center of mass of this system lies at the center of the Earth and we are perfectly justified in considering the Earth to be at rest.⁴

In considering the motion of a particle of mass m in a central force field, the first thing to note is that *the particle moves in a plane*. The proof of this statement is given by the following argument. At any instant in time the particle is in the plane defined by the position vector and the velocity vector. The only way the particle can move out of this plane is if its acceleration has a component perpendicular to the plane. But for a central force, the force (and hence the acceleration) lies along $\mathbf{r}$, and therefore there is no component of the acceleration perpendicular to the plane.

⁴ Recall the discussion of reduced mass, defined as $\mu = m_1 m_2 / (m_1 + m_2)$. See Equation (6.20). If the central force problem is treated in terms of the reduced mass, then no approximation is made. The relations obtained are the same as those derived here except that m is replaced by μ and $\mathbf{r}$ is the position of one body relative to the other rather than the distance from the center of mass.

Perhaps the most important aspect of the motion of a particle under the action of a central force is that the moving particle *has a constant angular momentum*. By definition, the angular momentum of a particle is $\mathbf{l} = \mathbf{r} \times \mathbf{p}$, where $\mathbf{r}$ is the position of the particle and $\mathbf{p}$ is its linear momentum. Recall from Section 7.2.2 that the time derivative of the angular momentum is

$$\frac{d\mathbf{l}}{dt} = \frac{d}{dt}(\mathbf{r} \times \mathbf{p}) = \mathbf{r} \times \mathbf{F}.$$

In the case of a central force, $\mathbf{F} = f(r)\hat{\mathbf{r}}$. Hence,

$$\begin{aligned}\frac{d\mathbf{l}}{dt} &= \mathbf{r} \times f(r)\hat{\mathbf{r}} = r\hat{\mathbf{r}} \times f(r)\hat{\mathbf{r}} \\ &= rf(r)(\hat{\mathbf{r}} \times \hat{\mathbf{r}}) = 0.\end{aligned}$$

Consequently, for a central force, the time derivative of the angular momentum is zero. This means that the angular momentum is constant. Also recall that the time rate of change of angular momentum is equal to the torque. A central force does not exert a torque on a particle.

These simple physical arguments lead us to an important conservation law concerning the motion of *any* particle under the action of *any* central force:

The angular momentum is constant.

This means that both the *magnitude* and the *direction* of the angular momentum are constant.

For a more sophisticated proof of the constancy of angular momentum, you can write the Lagrangian for a particle in a central force field. In polar coordinates the kinetic energy is $T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$ and the potential energy is $V = -\int_{r_0}^r f(r)dr$. The Lagrangian is

$$L = T - V = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + \int_{r_0}^r f(r)dr.$$

Since θ is ignorable, $\partial L/\partial\theta = \text{constant}$. That is,

$$\frac{\partial L}{\partial\dot{\theta}} = mr^2\dot{\theta} = l = \text{constant}.$$

We can use the constancy of angular momentum to generate another proof that a particle in a central force field moves in a plane. Assume the particle is at position $\mathbf{r}$ and has velocity $\mathbf{v}$. These two vectors define a plane and, of course, the particle lies in this plane. By the definition of the cross product, the vector $\mathbf{r} \times \mathbf{v}$ is perpendicular to the plane containing the vectors $\mathbf{r}$ and $\mathbf{v}$. But $\mathbf{l} = m\mathbf{r} \times \mathbf{v}$, so the vector $\mathbf{l}$ is perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{v}$. Because $\mathbf{l} = \text{constant}$, the perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{v}$ is constant. Therefore, the particle moves in a constant plane. See Figure 10.2.

Because the motion of the particle lies in a plane, two coordinates are sufficient to specify its position. These can be x and y or r and θ . The origin of the coordinate system is usually placed at the primary (assumed to be at rest).⁵

[Vector diagram: angular momentum vector $\mathbf{l}$ perpendicular to plane containing $\mathbf{r}$ and $\mathbf{v}$]

Figure 10.2 The angular momentum vector is perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{v}$. Because the angular momentum is constant, the plane is invariant.

To orient the coordinates, it is necessary to specify a fixed direction. Astronomers pick an imaginary line from the center of the Earth towards a position called “The First

Point in Aries” which is the position of the Sun at the vernal equinox.

[Diagram of orbit with angle θ and First Point in Aries symbol]

In Figure 10.3 the symbol Υ indicates the fixed line in space. In polar coordinates the angle θ is measured from Υ . In Cartesian coordinates one usually defines the x -axis along this fixed line. The y -axis is selected in the plane of the orbit and perpendicular

to the x -axis. The z -axis is then perpendicular to the orbit and along the angular momentum vector. For many problems it is safe to assume that this coordinate system is an inertial system.

Figure 10.3 The position of a planet referenced to "The First Point in Aries."

Five thousand years ago the position of the Sun at noon on the vernal equinox was in the constellation Aries (“the ram”). Because the Earth’s axis of rotation precesses with a period of about 25,000 years, the position of the Sun at noon on the vernal equinox has changed and it is presently in the constellation Pisces; within a few hundred years it will enter into the constellation Aquarius. However, the name “First Point in Aries” and the symbol Υ , representing ram’s horns, are still used to represent this arbitrary fixed line in space. It is amusing to note that due to the precession of the Earth’s axis, the positions of the zodiacal constellations have shifted, but astrologers still use the values of 5000 years ago. Thus, people born when the Sun was in the constellation Pisces think they are Aries, those who were born when the Sun was in Aries think they are Taurus, and so on. If astrology had any validity, this horrible mix-up in the zodiacal signs would be serious indeed!

⁵ In the language of astronomy, if one body is much more massive than the other, the massive body is called the “primary.”

Worked Example 10.1

A particle is in a circular orbit under the action of an inverse cubed attractive central force given by $f(r) = -k/r^3$. Obtain an expression for the angular momentum and show that it is constant.

Solution By Newton’s second law, $m\ddot{\mathbf{r}} = -f(r)\hat{\mathbf{r}} = -\frac{k}{r^3}\hat{\mathbf{r}}$. Recall from our discussion of plane polar coordinates in Chapter 2 that the acceleration in polar coordinates is

$$\ddot{\mathbf{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}}.$$

For a circular orbit, the magnitude r is constant, so $\dot{r} = 0$ and $\ddot{r} = 0$ so

$$m\ddot{\mathbf{r}} = m(-r\dot{\theta}^2\hat{\mathbf{r}} + r\ddot{\theta}\hat{\boldsymbol{\theta}}) = -f(r)\hat{\mathbf{r}}.$$

Therefore, for an inverse cube force law, $m\ddot{\mathbf{r}} = -f(r)\hat{\mathbf{r}}$ yields two relations, namely

$$-mr\dot{\theta}^2 = -f(r) = -\frac{k}{r^3},$$

and

$$mr\ddot{\theta} = 0.$$

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The second equation boils down to $\ddot{\theta} = 0$, telling us that the angular velocity, $\dot{\theta}$ is constant. The first equation yields

$$mr^4\dot{\theta}^2 = k.$$

But because $l = mr^2\dot{\theta}$ we see that the left-hand side is l^2/m and consequently

$$l^2 = mk,$$

and hence the angular momentum is constant. This result also follows from $r = \text{constant}$ and $\dot{\theta} = \text{constant}$.

Worked Example 10.2

Consider a particle in an attractive force field whose potential energy is of the form $V(r) = kr^{n+1}$. Show that for a periodic orbit the average kinetic energy is related to the average potential energy by

$$\langle T \rangle = \frac{n+1}{2} \langle V \rangle.$$

Apply to the gravitational force. (This is a special case of the virial theorem.)

Solution This problem asks us to determine the relationship between the average values of the kinetic and potential energy. The time average of any quantity (such as the kinetic energy) is defined as:

$$\langle T \rangle = \frac{1}{\tau} \int_0^\tau T(t) dt.$$

For this problem, it is reasonable to let τ be the orbit period. If a particle is in orbit, after one period it will have returned to its original location and have the same velocity as it did initially. Therefore, the quantity $G(t) = \mathbf{p} \cdot \mathbf{r}$ repeats periodically. Let us define the periodic function $G(t) = \mathbf{p} \cdot \mathbf{r}$. Consider the time average of the derivative of G with respect to time:

$$\left\langle \frac{dG}{dt} \right\rangle = \frac{1}{\tau} \int_0^\tau \frac{dG}{dt} dt = \frac{1}{\tau} \int_0^\tau dG = \frac{1}{\tau} [G(\tau) - G(0)] = 0.$$

But

$$\begin{aligned} \frac{dG}{dt} &= \frac{d}{dt}(\mathbf{p} \cdot \mathbf{r}) = \mathbf{p} \cdot \dot{\mathbf{r}} + \dot{\mathbf{p}} \cdot \mathbf{r} \\ &= m\mathbf{v} \cdot \mathbf{v} + \mathbf{F} \cdot \mathbf{r} = 2T - \nabla V \cdot \mathbf{r} \\ &= 2T - \frac{dV}{dr} r, \end{aligned}$$

because $\nabla V \cdot \mathbf{r} = \frac{dV(r)}{dr} r$. Therefore,

$$\left\langle \frac{dG}{dt} \right\rangle = \frac{1}{\tau} \int_0^\tau \left(2T - \frac{dV}{dr} r \right) dt = 0,$$

and

$$\begin{aligned} 2\langle T \rangle - \left\langle \frac{dV}{dr} r \right\rangle &= 0, \\ \langle T \rangle &= \frac{1}{2} \left\langle \frac{dV}{dr} r \right\rangle. \end{aligned}$$

Since $V = kr^{n+1}$ we have

$$\frac{dV}{dr}r = (n+1)kr^n r = (n+1)kr^{n+1} = (n+1)V,$$

and

$$\left\langle \frac{dV}{dr}r \right\rangle = (n+1)\langle V \rangle,$$

so

$$\langle T \rangle = \frac{n+1}{2}\langle V \rangle.$$

The gravitational potential has the form $V = -\frac{k}{r}$ so $n = -2$ and

$$\langle T \rangle = -\frac{1}{2}\langle V \rangle.$$

This relationship between kinetic and potential energy is useful when solving orbital mechanics problems. It might be mentioned that the virial theorem has important applications in thermodynamics and statistical mechanics.

Exercise 10.1

Use $\mathbf{F} = -\nabla V$ to obtain an expression for the Lennard-Jones force. Determine the value of r where the force changes from attractive to repulsive. Answer: $\mathbf{F} = -\hat{\mathbf{r}} \left(\frac{6a}{r^7} - \frac{12b}{r^{13}} \right)$.

Exercise 10.2

Use $V = -\int_{r_0}^r F(r)dr$ to obtain the potential energy for the gravitational force, the electrostatic force, and the force exerted by a spring. Select r_0 appropriately.

Exercise 10.3

A planet of mass m is in a circular orbit of radius r_0 about a star of mass M . Determine its angular momentum. Answer: $l = m\sqrt{GMr_0}$.

Exercise 10.4

Assume that at some initial moment the radius vector $\mathbf{r}$ and the velocity vector $\mathbf{v}$ are perpendicular to one another – however, the angle between them is changing with time. Explain why the two vectors cannot ever be parallel to one another. Answer: Because angular momentum is constant.

10.4 The Equation of Motion

Our next task is to determine the equation of motion for a planet. We will do this in two ways: first by writing down Newton's second law, and second by applying the Lagrangian technique.

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The force acting on a particle (planet) of mass m is given by Newton's law of universal gravitation

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}},$$

where M is the mass of the larger attracting body, assumed fixed at the origin of coordinates. The equation of motion of the satellite (mass m) is

$$m\ddot{\mathbf{r}} = -G\frac{Mm}{r^2}\hat{\mathbf{r}},$$

or

$$\ddot{\mathbf{r}} = -\frac{GM}{r^2}\hat{\mathbf{r}}. \quad (10.1)$$

We evaluated the acceleration $\ddot{\mathbf{r}}$ in polar coordinates in Chapter 2 – see Equation (2.14)

– obtaining

$$\ddot{\mathbf{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}.$$

Substituting this expression into Equation (10.1), yields

$$(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta} = -\frac{GM}{r^2}\hat{\mathbf{r}}.$$

Separating the radial and angular components leads to the following two scalar equations

$$\begin{aligned}\ddot{r} - r\dot{\theta}^2 &= -\frac{GM}{r^2}, \\ r\ddot{\theta} + 2\dot{r}\dot{\theta} &= 0.\end{aligned}\tag{10.2}$$

These equations are a pair of *coupled* second-order ordinary differential equations. The second equation, the “ θ -equation,” is easy to analyze by going back to the definition of angular momentum and recalling that

$$\mathbf{l} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times m\mathbf{v}.$$

The velocity in plane polar coordinates is $\mathbf{v} = \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\theta}$, so

$$\mathbf{l} = mr\hat{\mathbf{r}} \times (\dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\theta}).$$

Now carry out the cross products, using the facts that $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = 0$ and $\hat{\mathbf{r}} \times \hat{\theta} = \hat{\mathbf{k}}$, where $\hat{\mathbf{k}}$ is perpendicular to the plane of motion. You obtain

$$\mathbf{l} = mr(r\dot{\theta})\hat{\mathbf{k}} = mr^2\dot{\theta}\hat{\mathbf{k}}.\tag{10.3}$$

This, in itself, is a useful equation. But it becomes even more useful if you take its time derivative,

$$\frac{d\mathbf{l}}{dt} = \frac{d}{dt}(mr^2\dot{\theta})\hat{\mathbf{k}}.$$

For central forces, the angular momentum is constant, so $d\mathbf{l}/dt = 0$ and the left-hand side of this equation is zero. Therefore,

$$\begin{aligned}0 &= \frac{d}{dt}(mr^2\dot{\theta}) = m(2r\dot{r}\dot{\theta} + r^2\ddot{\theta}), \\ &= mr(2\dot{r}\dot{\theta} + r\ddot{\theta}).\end{aligned}$$

Since neither m nor r is equal to zero, this implies that

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0.$$

But this is just the θ equation! Therefore, we see that the θ equation of motion, the second of Equations (10.2), is a statement that angular momentum is constant. Consequently, that equation can be replaced by the equivalent relation

$$l = mr^2\dot{\theta} = \text{constant}.$$

This equation, in turn, gives a nice expression for $\dot{\theta}$, namely,

$$\dot{\theta} = \frac{l}{mr^2}. \quad (10.4)$$

Replacing $\dot{\theta}$ in the r equation, the first of Equations (10.2), by expression (10.4) yields the following equation for the radial motion of mass m :

$$\ddot{r} - \frac{l^2}{m^2r^3} = -\frac{GM}{r^2}. \quad (10.5)$$

This equation involves only r . Thus, conservation of angular momentum *decouples* the equations of motion. The equation has only one variable (r), so it is often called a “one-dimensional equation.” But you should always keep in mind that the motion takes place in two dimensions. Once Equation (10.5) has been solved for $r = r(t)$, you can use it in Equation (10.4) to determine $\theta = \theta(t)$.

We shall now use the Lagrangian technique to determine the equations of motion. (We had better obtain the same result!)

As you know, the Lagrangian is $L = T - V$. In this problem V is the potential energy for a particle of mass m attracted gravitationally to a body of mass M . According to Equation (9.6) this is

$$V(r) = -\frac{GMm}{r}.$$

In polar coordinates, the kinetic energy ($\frac{1}{2}mv^2 = \frac{1}{2}m\mathbf{v} \cdot \mathbf{v}$) is

$$T = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2.$$

Therefore the Lagrangian is

$$L = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + \frac{GMm}{r}.$$

Recall that the Lagrange equations of motion have the form

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0,$$

where the q_i 's are now r and θ . So we have two equations, namely,

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{r}} - \frac{\partial L}{\partial r} = 0,$$

and

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0.$$
